6G Gigantic MIMO: From Antenna Abundance to Antenna Intelligence

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What I want you to remember

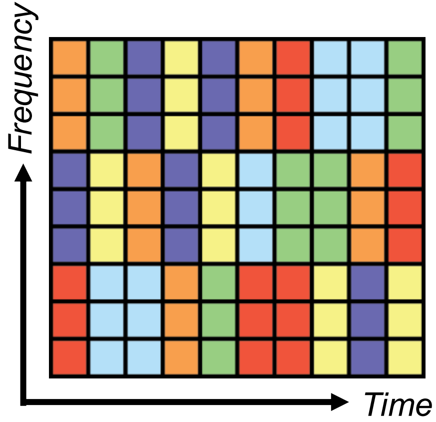
- Larger antenna arrays are essential for 6G
- We should rethink the array geometry
- The role that *movable antennas* might play

Evolution in Multiple Access

1. Sectorization and frequency reuse

2. Multi-user MIMO (multiple-input multiple-output)

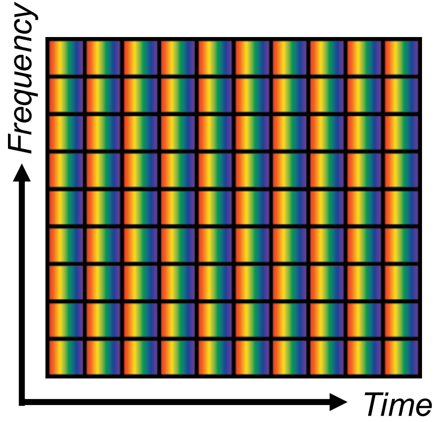
Time-frequency scheduling



Beamforming:
Higher SNR
Channel hardening

Spatial multiplexing:
Higher cell capacity

User-separation by beamforming

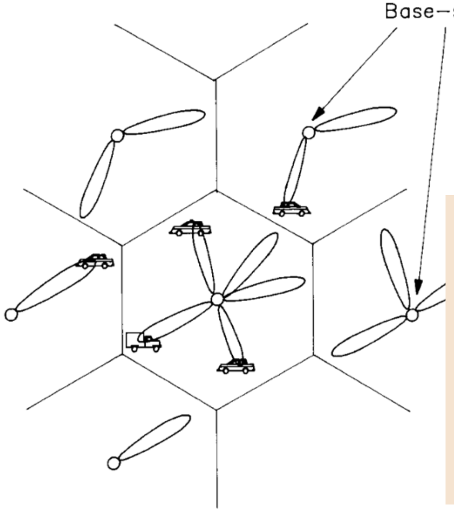


From **SDMA** to **Massive MIMO**

Space-division multiple access

Concept from late 80s, early 90s

Information theory in 00s



Assum

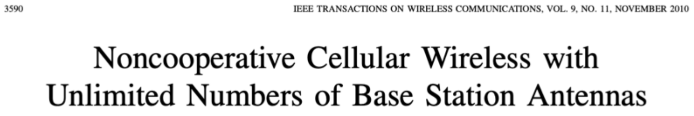
M

$M = $ #a

$K = $ #

Reference: Swales, Beach, Edwards, McG
Performance Enhancement of Multibeam A
Base-Station Antennas for Cellular…," IEEE

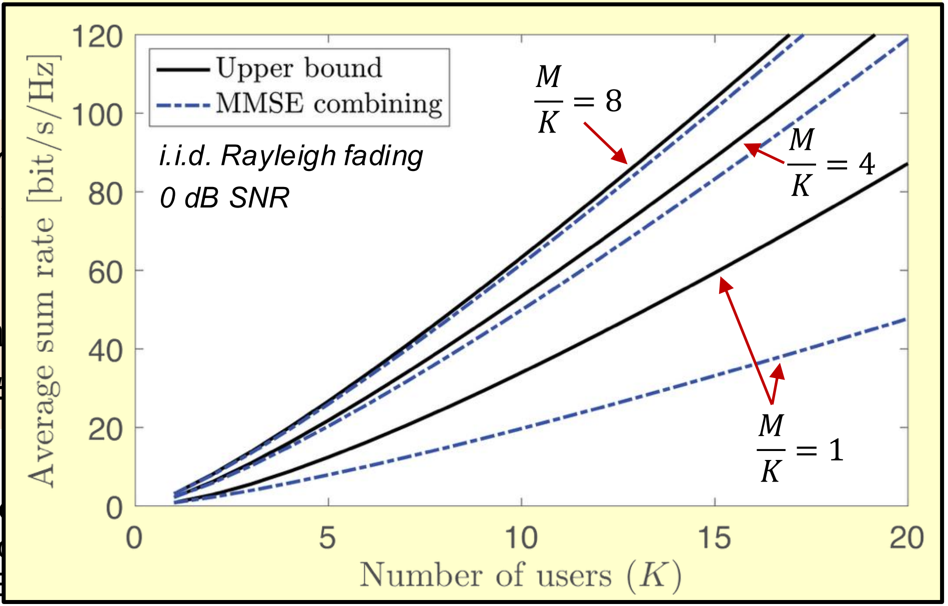
Marzetta's "Massive" MIMO (2010)


3590 IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, VOL. 9, NO. 11, NOVEMBER 2010

Noncooperative Cellular Wireless with
Unlimited Numbers of Base Station Antennas

Assumption
$M \gg K$

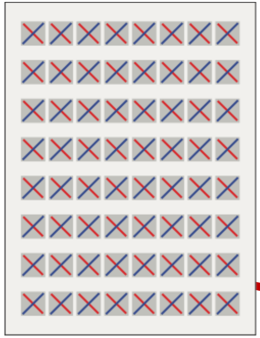
Abundance of antennas → "Favorable propagation"



Massive MIMO in 5G

n78 band: 3.3-3.8 GHz

100 MHz bandwidth



64 antennas

One device: 710 Mbps (average)

MU-MIMO with 8 user devices

- 16 spatial layers, 2 layers/user
- 650 Mbps per device (average)
- 5.2 Gbps in total (52 bit/s/Hz)

$$\frac{M}{K} = 4$$

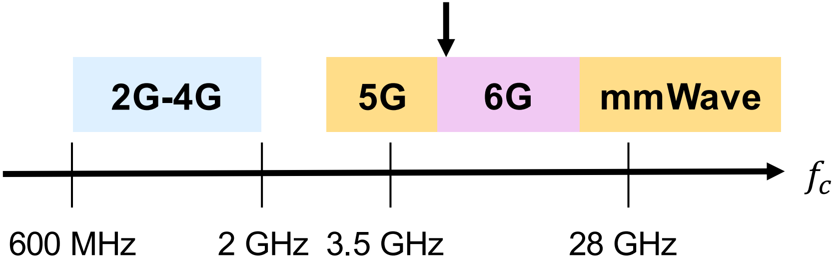
4 antennas

Reference: Signals Research Group, "The Industry's First Independent Benchmark Study of 5G NR MU-MIMO," 2020

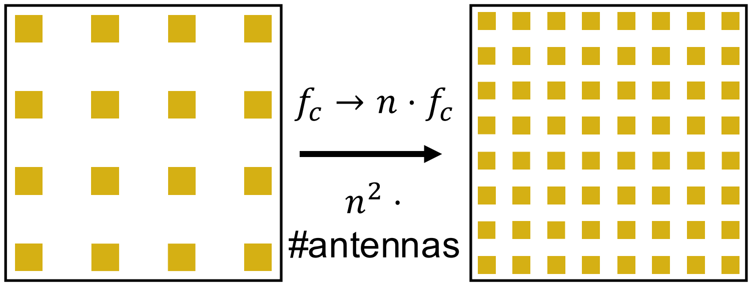
From **Massive** to **Gigantic MIMO** in **6G**

Spectrum

Upper mid-band
7.1-8.4 GHz



Antennas fitting in the aperture



Example: $(7.8/3.5)^2 \approx 5$

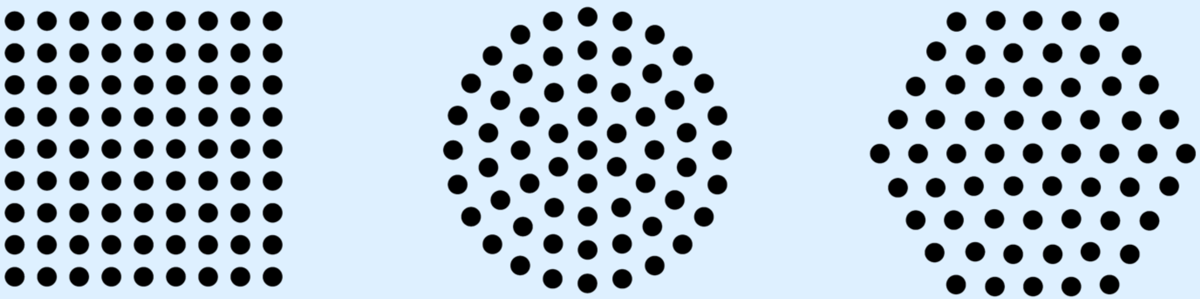
4G		**5G**		**6G**
20 MHz	$5\times$ →	100 MHz	$2\times$ →	200 MHz
4 antennas	$16\times$ →	64 antennas	$5\times$ →	320 antennas

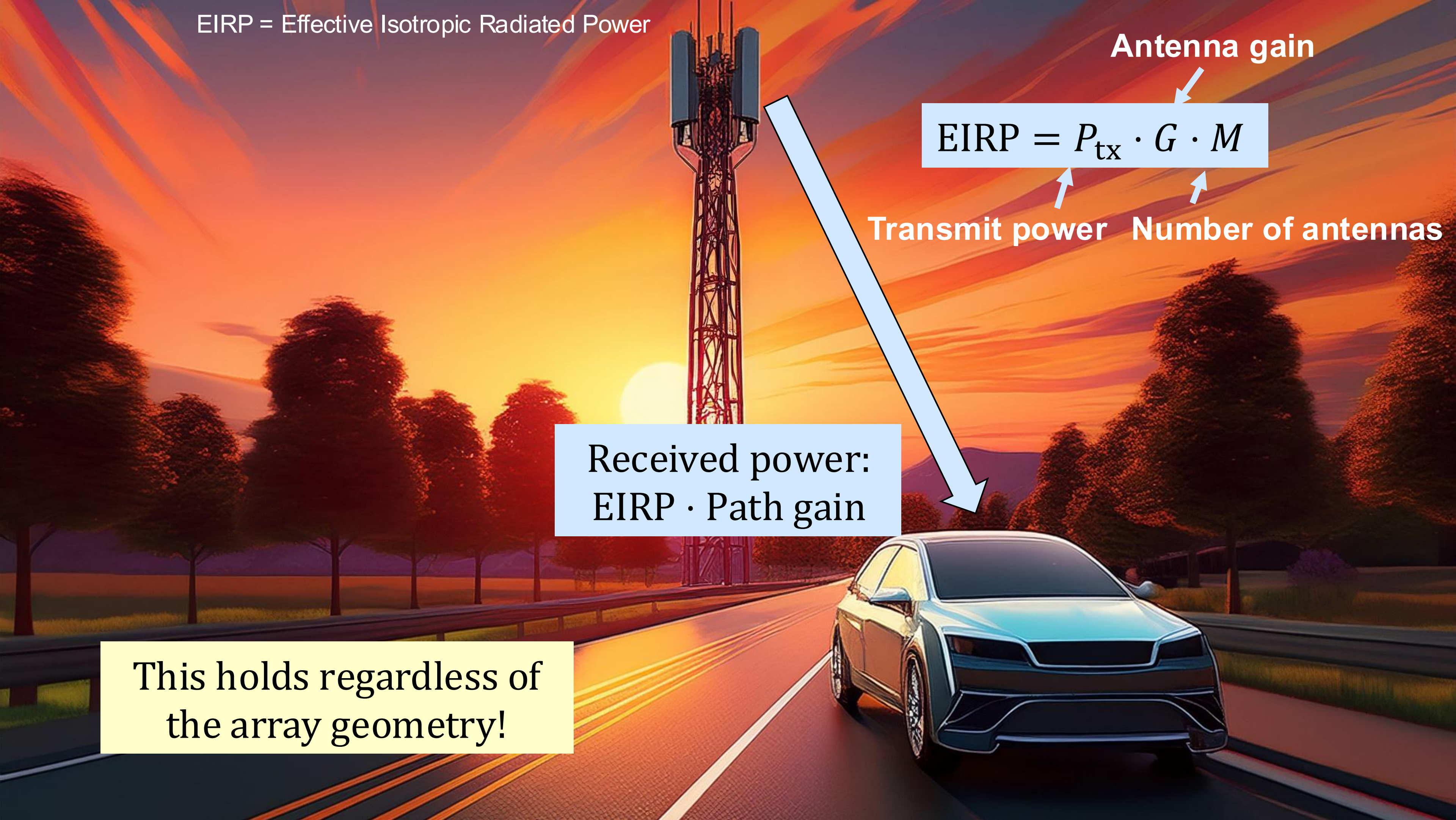
Capacity growth mainly achieved by ***bigger MIMO***!

Challenge: Very many antennas if we keep $M \gg K$

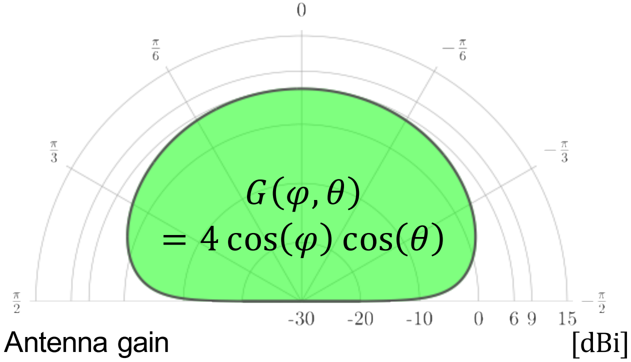
Can we avoid the antenna abundance by smarter array design?

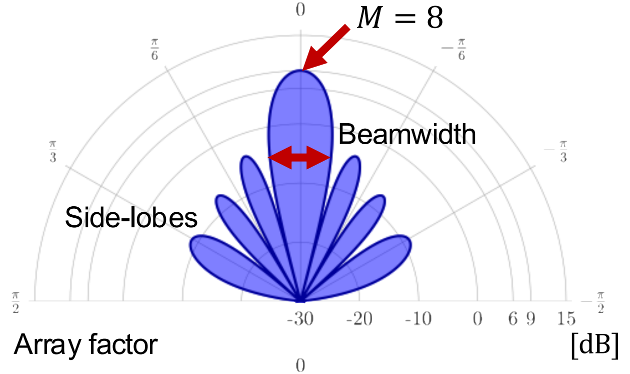
WHAT IS A GOOD ANTENNA ARRAY GEOMETRY?

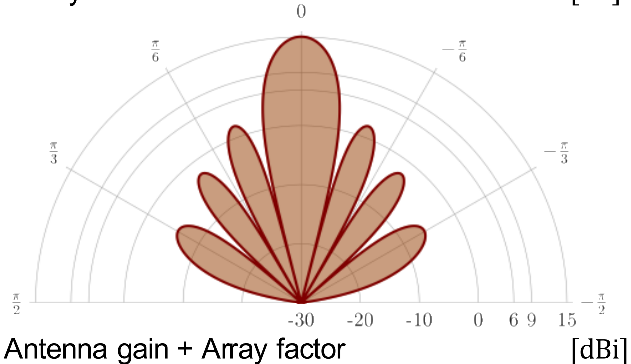


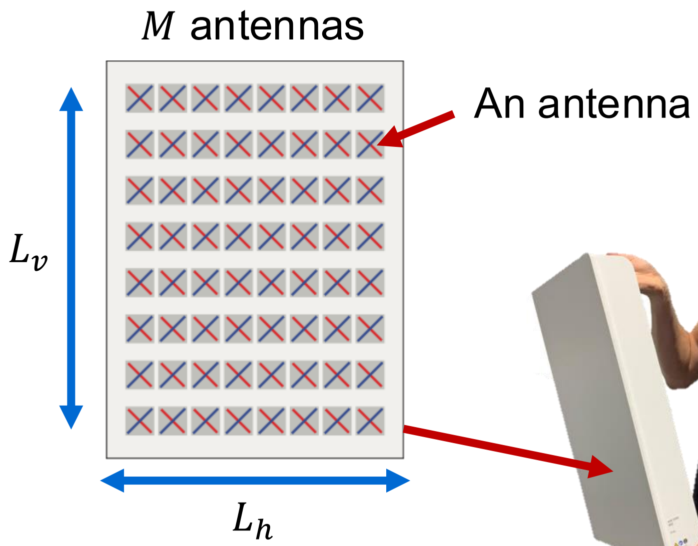

EIRP = Effective Isotropic Radiated Power
Antenna gain
$\mathrm{EIRP} = P_{\mathrm{tx}} \cdot G \cdot M$
Transmit power
Number of antennas
Received power:
$\mathrm{EIRP} \cdot$ Path gain
This holds regardless of the array geometry!

Key Properties of an Antenna Array





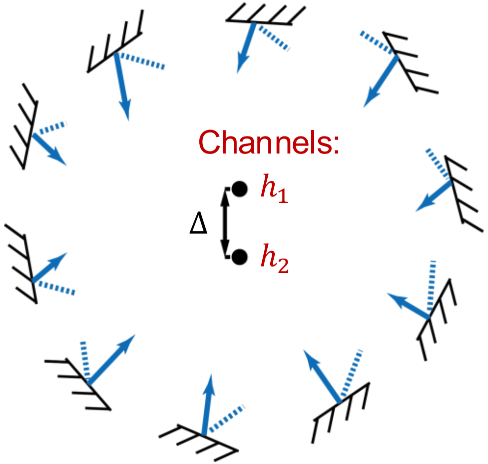




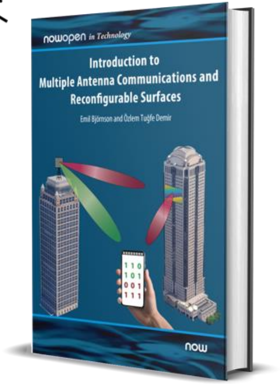
1. Antenna gain function: $G(\varphi, \theta)$
2. Beamforming gain: M
3. Beamwidth (first null): $\approx \frac{2\lambda}{L_h}, \frac{2\lambda}{L_v}$ rad
4. Side-lobe pattern: Array geometry

A **Classical Motivation** for $\lambda/2$-Spacing

Isotropic 3D scattering environment



$$E\{h_1 h_2^*\} = \cdots = \beta \cdot \mathrm{sinc}\left(\frac{\Delta}{\lambda/2}\right)$$



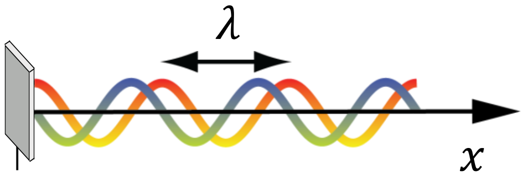
Nulls at $\Delta = \frac{\lambda}{2} \cdot \text{integer}$ (except 0)

Place antennas $\lambda/2$ apart!

Another **Classical Motivations** for $\lambda/2$-Spacing

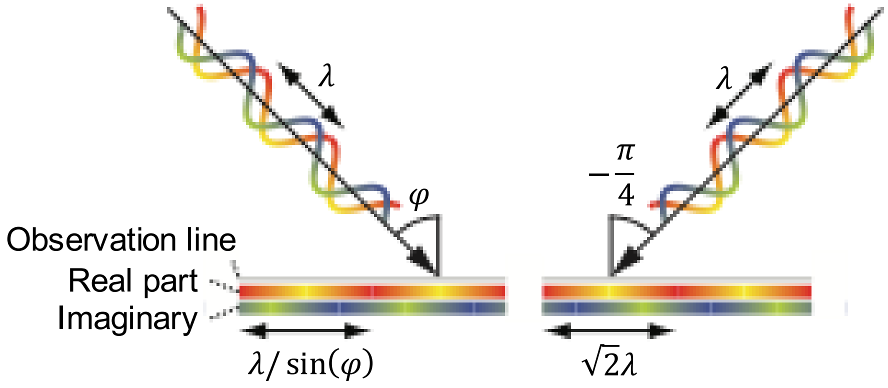
Data

Signal: $s \cdot e^{-\frac{j2\pi x}{\lambda}} \cdot e^{\frac{j2\pi ct}{\lambda}}$



Nominal spatial frequency: $\frac{1}{\lambda}$

Sampling of plane waves



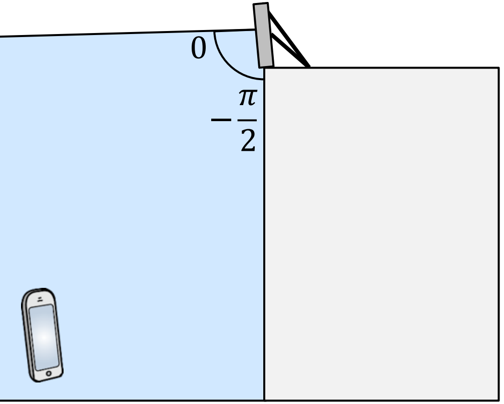
Arbitrary plane waves: $\varphi \in [-\pi/2, \pi/2]$

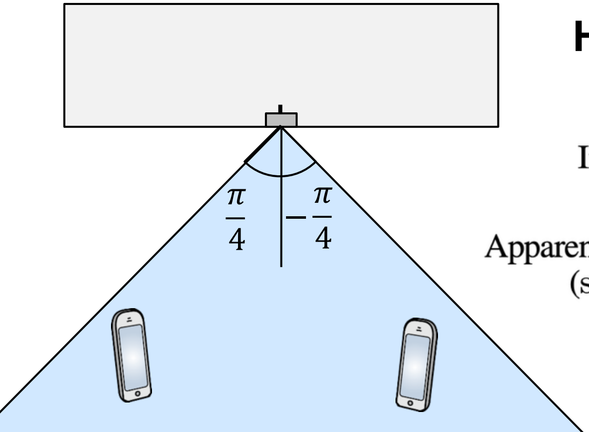
Spatial frequencies: $\sin(\varphi)/\lambda \in [-1/\lambda, 1/\lambda]$

Spatial bandwidth $2/\lambda$

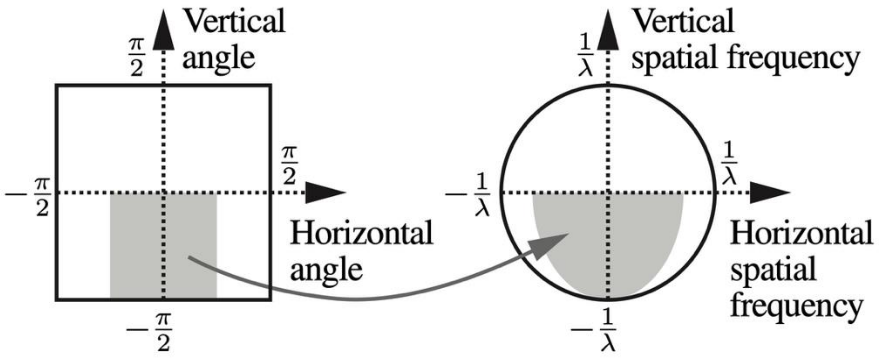
Sampling theorem (complex signal):
Place antennas to take samples $\lambda/2$ apart!

Practical Deployments are Different

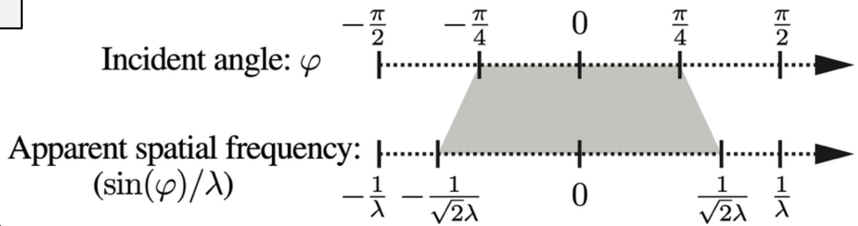




Joint horizontal and vertical coverage:



Horizontal coverage sector:



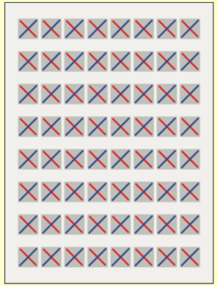
Place antennas $\sqrt{2}\lambda/2$ apart!

Use spacing:

$\sqrt{2}\lambda/2$ horizontally

λ vertically

Vertical spacing 0.7λ

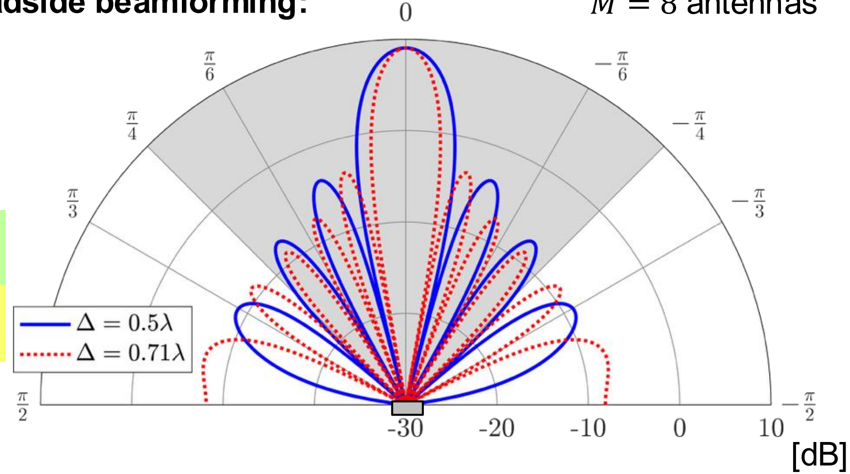


64 antennas

Sparse Uniform Arrays:
What is the Catch?

Broadside beamforming:

$M = 8$ antennas



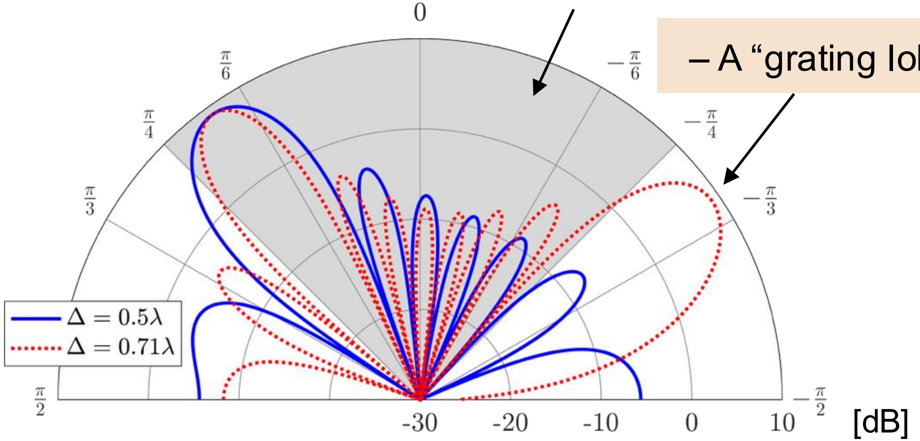
+ Narrower beam!

Different sidelobes

Beamforming toward $\pi/5$:

Coverage region

– A "grating lobe"



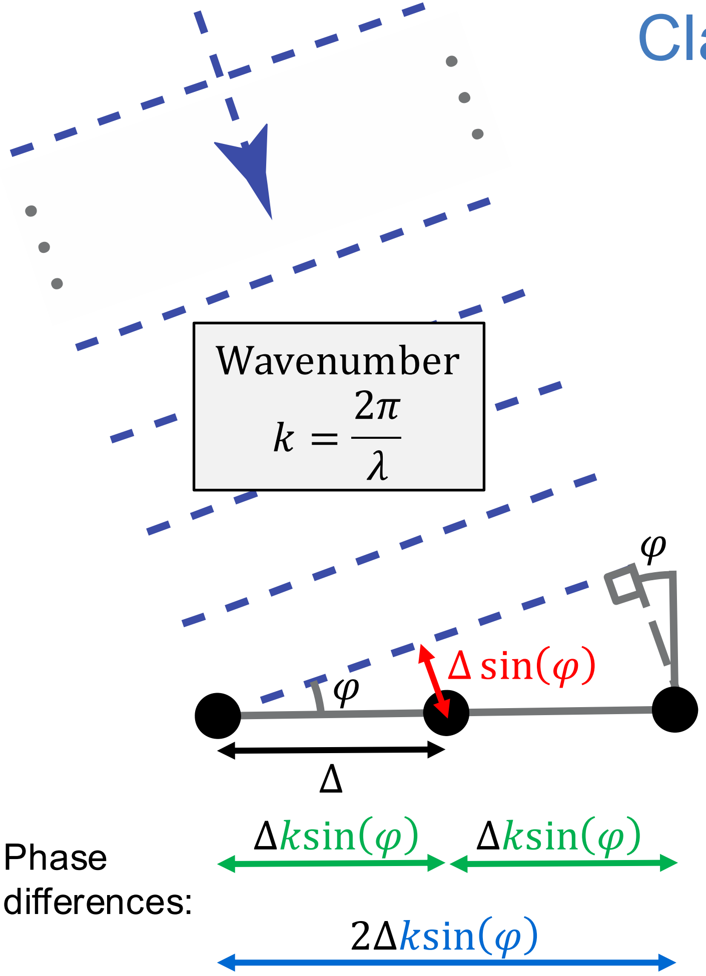
Are grating lobes
a **real issue?**

Yes, for localization and
interference to others

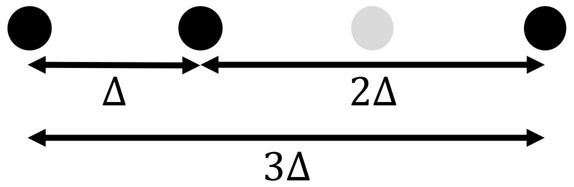
But…

Leads to better beam resolution
in the coverage region

Classical **Structured Non-Uniform** Arrays



Minimum-redundancy array (MRA):



Three unique separations:

Maximum number of phase differences!

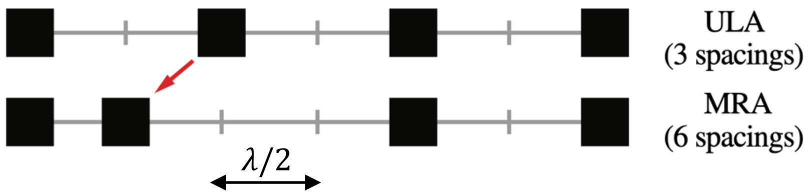
M-antennas arrays can generate up to $\binom{M}{2} = \frac{M(M-1)}{2}$ phase differences

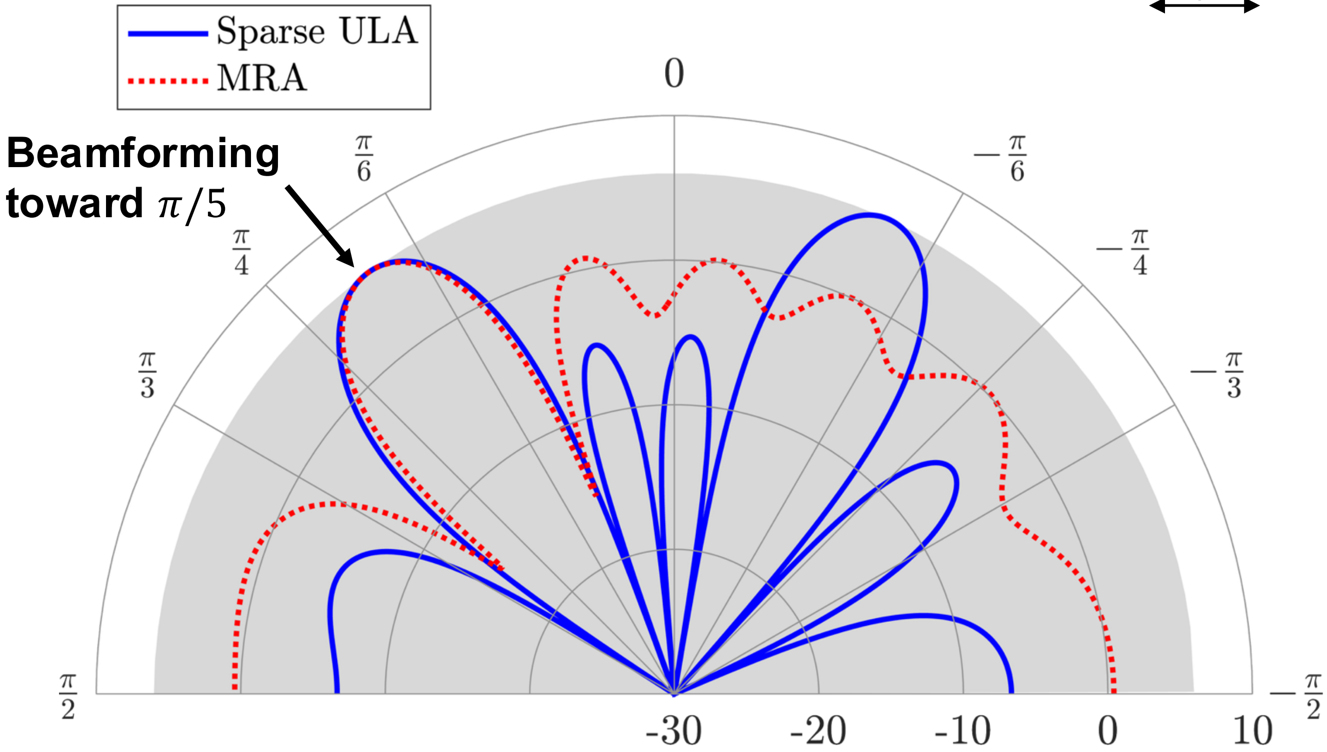
▼

Detect $\approx \frac{M^2}{2}$ sources with M antennas

Reference: Moffet, “Minimum-redundancy linear arrays,” IEEE TAP, 1968

What Happens to the Beam Pattern?





Are Non-Uniform Arrays **Useful for Communications**?

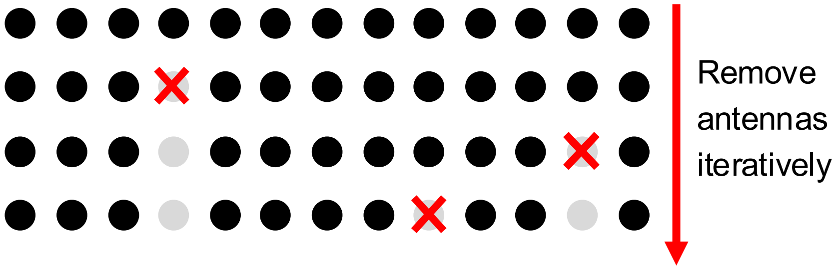
Channel estimation: Yes

For sparse channels (identify $M^2/2$ paths)

Beamformed data transmission: Unsure

M-length channels: At most M orthogonal beams
Mainlobe size ∝ 1/(array length)

Approach 1: **Thinned array**



Maximize communication performance

Example: Maintain average sum rate

Reference: Skolnik, Sherman, "Planar arrays with unequally spaced elements," Radio and Electronic Engineer, vol. 28, pp. 173–184, 1964.

Are Non-Uniform Arrays **Useful for Communications**?

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For sparse channels (identify $M^2/2$ paths)

Beamformed data transmission: Unsure

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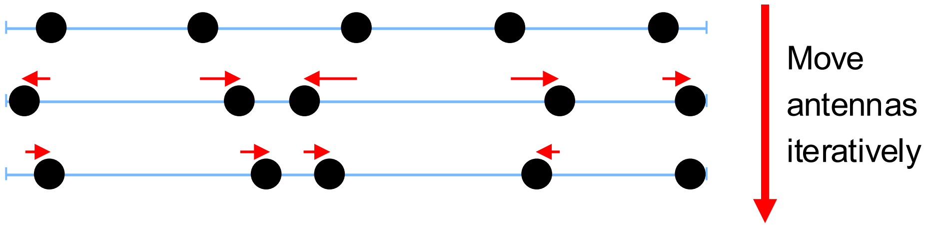
Mainlobe size ∝ 1/(array length)

Approach 2: **Pre-optimized Irregular Array (PIA)**

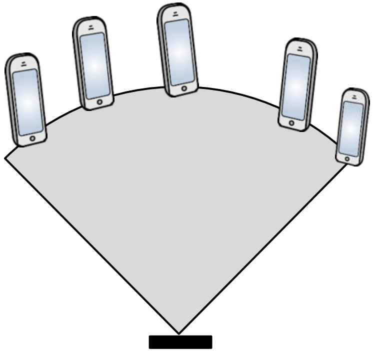


Use *particle swarm optimization (PSO)*
or similar method

Arbitrary spacings, but $\geq \lambda/2$ to avoid coupling

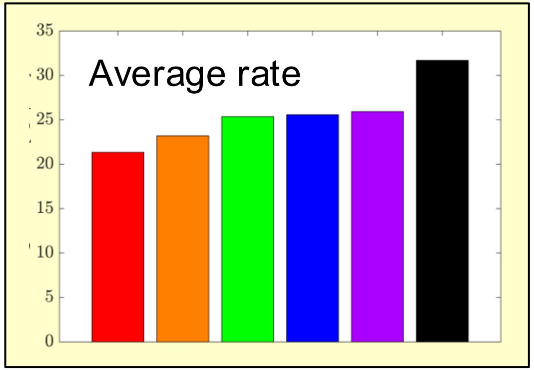
Reference: Irshad, Kosasih, Petrov, Björnson, "Pre-Optimized Irregular Arrays versus Movable Antennas in Multi-User MIMO Systems," IEEE WCL, 2025

Uplink Example: $M = 8$ antennas, $K = 5$ users



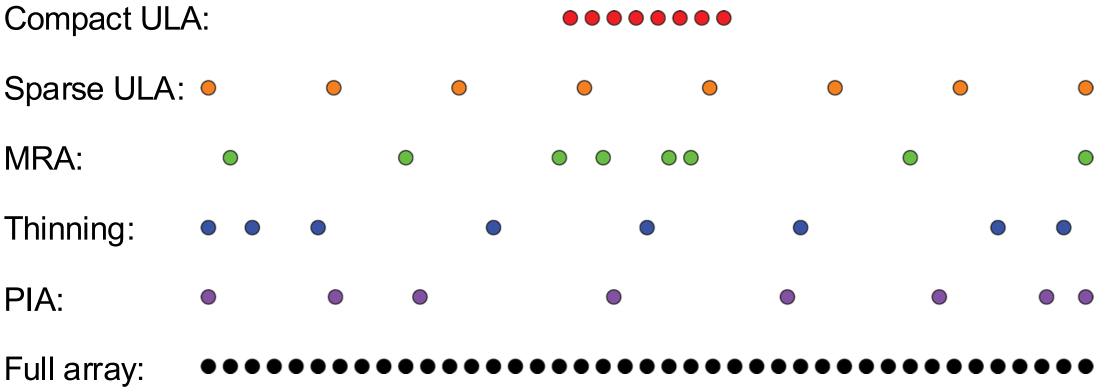
Non-uniform arrays are best!

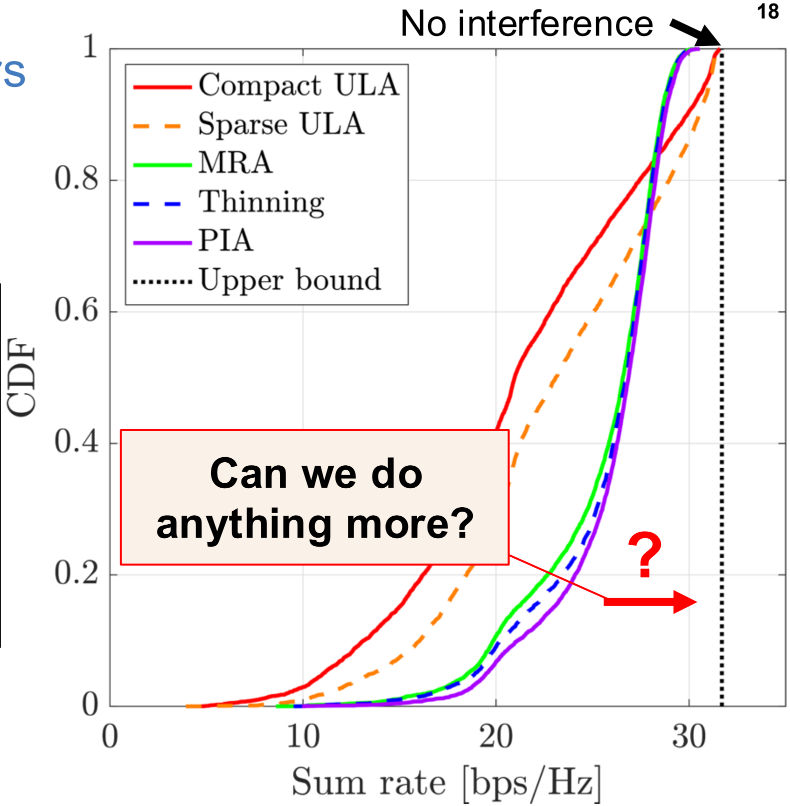
PIA a little better than MRA/thinning



Linear aperture: Length 20λ

- Line-of-sight, 10 dB SNR
- Random angles (10^4 user drops)

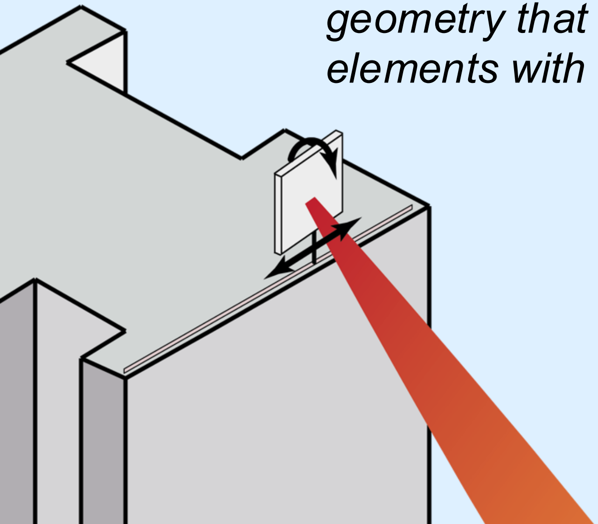




REAL-TIME MOVABLE ANTENNAS: *IS THAT A THING?*

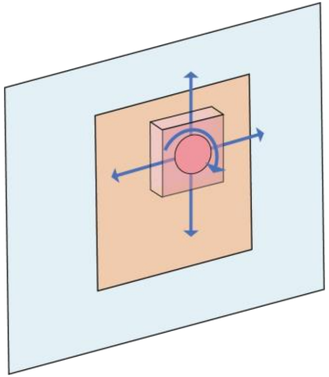
“…for a fixed array, there are certain things that cannot be done in the digital domain. […] To deal with such problems, an interesting avenue for future research is the study of arrays with flexible geometry that could optimally position themselves or their elements with some degrees of freedom.”

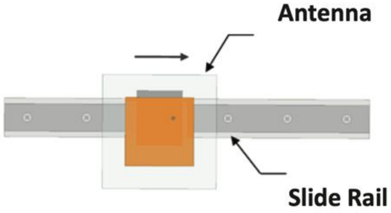
Björnson, Sanguinetti, Wymeersch, Hoydis, Marzetta, “Massive MIMO is a Reality—What is Next?”, DSP, 2019

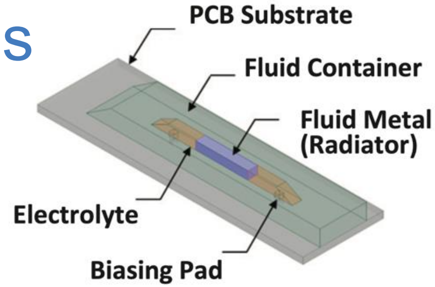


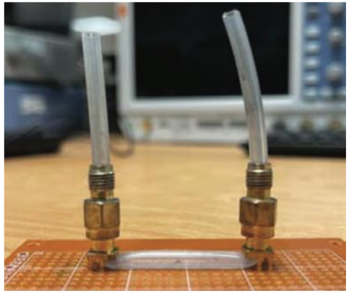
Hardware for Movable Antennas

Element-level

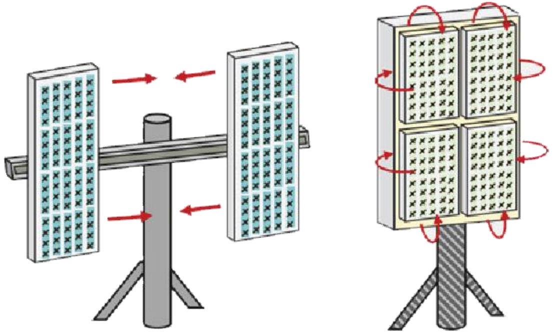








Array-level



Hardware-based terminologies

1999: MEMS reconfigurable antenna

2007: Mechanically movable antenna

2008: Electrolytic fluid antenna

MIMO-related concepts

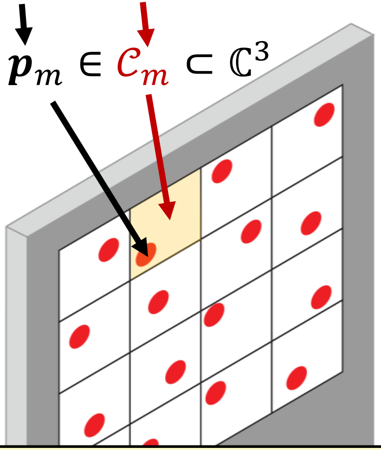
2019: Intelligent Massive MIMO

2020: Fluid antenna system

2022: Movable antenna system

Original sources cited in: Ning, Yang, Wu, Wang, Mei, Yuen, Björnson, “Antenna-Enhanced Wireless Communications: General Architectures and Implementation Methods,” IEEE Wireless Communications, 2025.

Optimizing Antenna Positions



Problem formulation:

$$\underset{\mathbf{p}_1,\ldots,\mathbf{p}_M}{\text{maximize}} \quad R_\Sigma(\mathbf{p}_1,\ldots,\mathbf{p}_M) \quad \longleftarrow \text{Sum rate (or another metric)}$$

$$\text{subject to} \quad \mathbf{p}_m \in \mathcal{C}_m,\ m = 1,\ldots,M \quad \longleftarrow \text{Movable area}$$

$$\|\mathbf{p}_m - \mathbf{p}_j\| \geq \lambda/2,\ m \neq j \quad \longleftarrow \text{Avoid mutual coupling}$$

Use parametrized channel:

$$\boldsymbol{h} = \sum_i c_i \boldsymbol{a}(\mathbf{p}_1,\ldots,\mathbf{p}_M,\varphi_i,\theta_i,)$$

Coefficient Array response Angles

Key challenges:

1. Acquire channel knowledge (*evaluate many* $R_\Sigma(\mathbf{p}_1,\ldots,\mathbf{p}_M)$)
2. Calculate the desired antenna locations (*solve problem above*)
3. Move the antennas quickly (*faster than channel coherence*)

PSO or something else

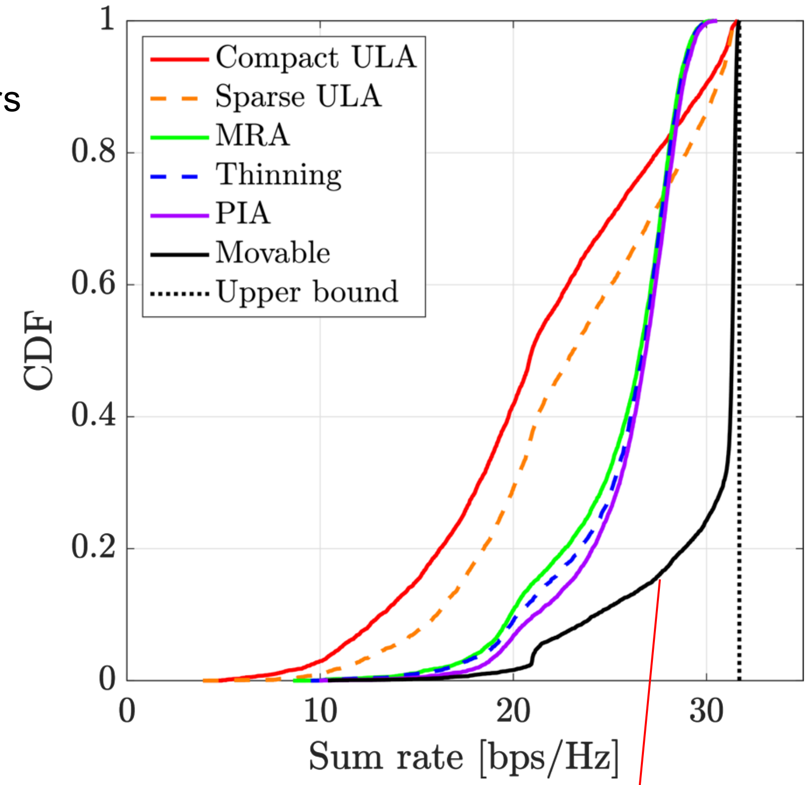
No real-time experiments?

Reference: Irshad, Kosasih, Petrov, Björnson, "Pre-Optimized Irregular Arrays versus Movable Antennas in Multi-User MIMO Systems," 2025

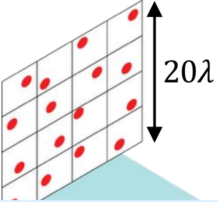
Uplink Example (continued)

Linear aperture: Length 20λ

$M = 8$ antennas, $K = 5$ users



Movable antennas achieve upper bound in 75% of setups



Wideband Example

$M = 16, K = 10$

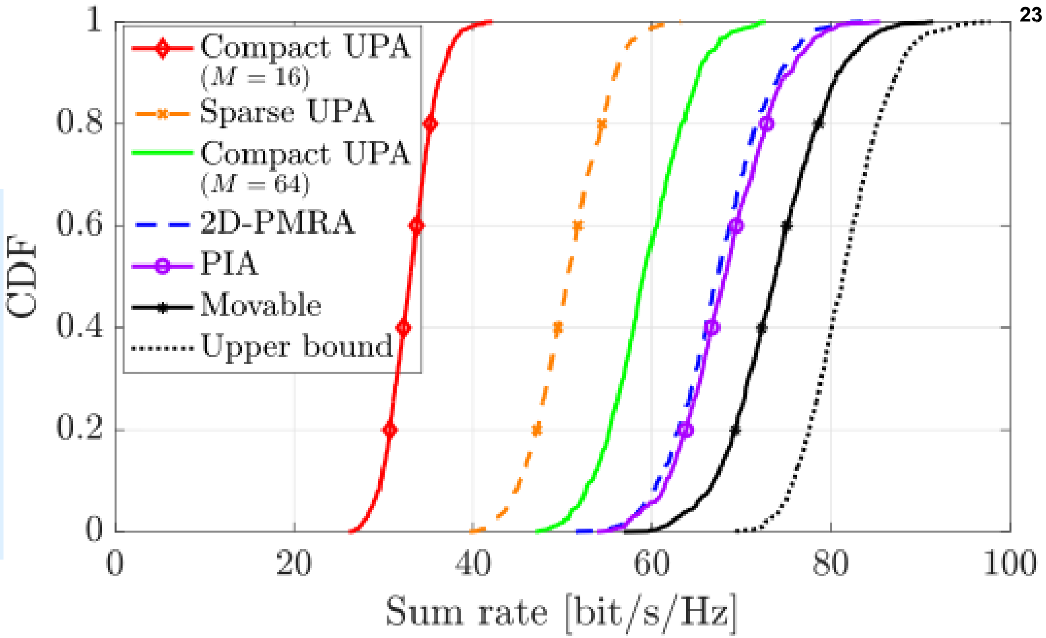
Observations

Non-uniform sparse arrays perform best

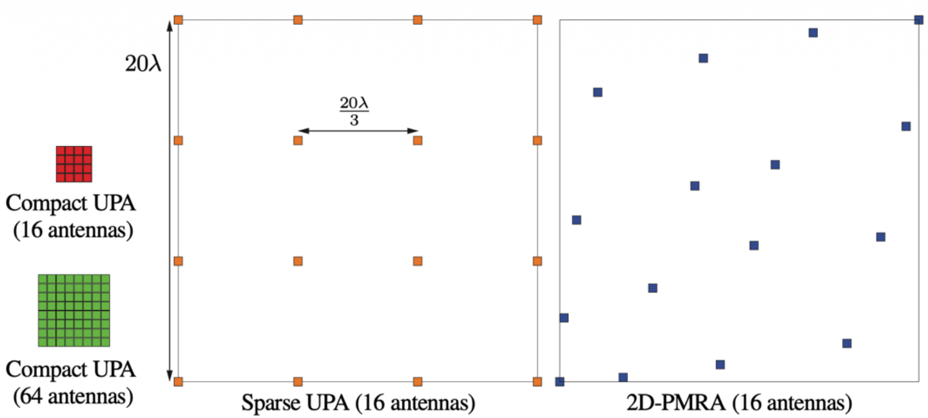
Enable reduced antenna number!

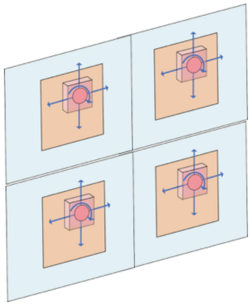
Movable antennas: Minor gain

Different desired locations across frequency



Benchmark schemes





Movable antennas

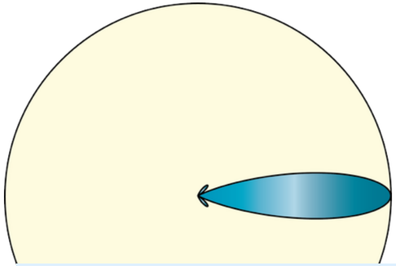
Maintaining the EIRP (Effective Isotropic Radiated Power)

Antenna gain

$$\mathrm{EIRP} = P_{\mathrm{tx}} \cdot G \cdot M$$

Transmit power

Number of antennas

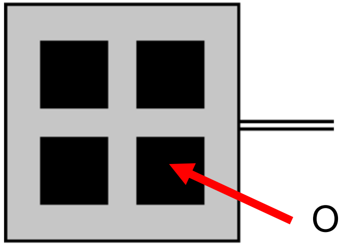


Fewer antennas

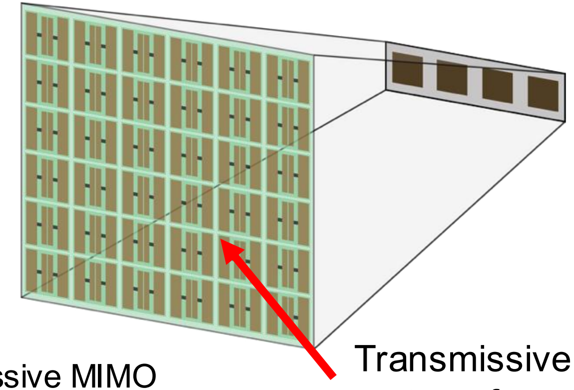
Larger gain to maintain EIRP

Option 1: Antennas with multiple elements

One "antenna port"



Option 2: RIS-Integrated antenna array



Reference: Demir, Björnson, "User-Centric Cell-Free Massive MIMO With RIS-Integrated Antenna Arrays," IEEE SPAWC 2024.

Summary: **From Antenna Abundance to Antenna Intelligence**

We need more spatial multiplexing in 6G bands

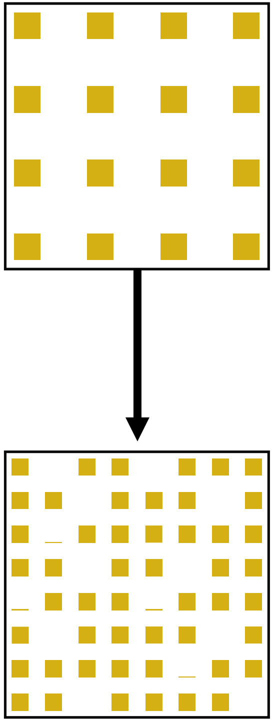
- Larger M (antennas) and K (users)

We can scale M more slowly (reduced M/K) using

- Non-uniform array tailored for the coverage area
- Tweak antenna gains to maintain EIRP

Real-time movable antennas

- Only useful in LOS scenarios
- Gains are only large compared to “wrong” benchmarks



A special thanks to

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OPEN CALL ARTICLE

From Antenna Abundance to Antenna Intelligence in 6G Gigantic MIMO Systems

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Abstract

Current cellular systems achieve high spectral efficiency through Massive MIMO, which leverages an abundance of antennas to create favorable propagation conditions for multiuser spatial multiplexing. Looking towards future networks, the extrapolation of this paradigm leads to systems with many hundreds of antennas per base station, raising concerns regarding hardware complexity, cost, and power consumption. This article suggests more intelligent array designs that reduce the need for excessive antenna numbers. We revisit classical uniform array design principles and explain how their uniform spatial sampling leads to unnecessary redundancies in practical deployment scenarios. By exploiting non-uniform sparse arrays with site-specific antenna placements—based on either pre-optimized irregular arrays or real-time movable antennas—we demonstrate how superior multiuser MIMO performance can be achieved with far fewer antennas. These principles are inspired by previous works on wireless localization. We explain and demonstrate numerically how these concepts can be adapted for communications to improve the average sum rate and similar metrics. The results suggest a paradigm shift for future antenna array design, where antenna intelligence replaces sheer antenna count. This opens new opportunities for efficient, adaptable, and sustainable Gigantic MIMO systems.

communicate with them simultaneously over the entire frequency band. If the beams are designed based on channel state information (CSI) to optimally balance between signal amplification and co-user interference suppression, SDMA enables a multiplexing gain of $\min(M,K)$. This means that the sum rate (bit/s/Hz) is proportional to the multiplexing gain when the signal-to-noise ratio (SNR) is high and greatly surpasses the sum rate achieved with traditional orthogonal access schemes.

SDMA was conceived as early as the 1990s [1], but it was then commonly assumed that $M \approx K$ because the multiplexing gain is not improved when one parameter deviates from the other. In 2010, Marzetta challenged this assumption with his *Massive MIMO* concept where $M \gg K$ [2]. Fig. 1(a) shows simulation results that highlight the main motivation for having this abundance of antennas. The figure shows how the sum rate grows with the number of users when M/K equals 1, 4, or 8. The sum rate is averaged over many channel realizations, generated using independent and identically distributed (iid) Rayleigh fading. Dashed curves indicate the performance achieved by regularized zero-forcing (RZF) beamforming based on perfect CSI, while solid curves indicate the theoretical maximum if co-user interference is neglected. There is a significant gap between the solid and dashed curves when $M/K = 1$, but the gap narrows when M/K increases

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